Project Module 2: Explainability System Design **

Objective:

Build an interpretable ML system and analyze how and why it makes decisions.

You will:

- Use the same dataset and preprocessing modules you chose in **Module 1**
- Reuse your previous code starter notebook
- Focus on understanding decisions, not improving accuracy

Deliverables:

1. Notebook containing full code
2. Technical report responding to the module questions. **Every answer must reference specific notebook outputs (figures, tables, or code blocks)**

** You are encouraged to use Large Language Models (LLMs) to support your thinking and learning. If you do so, you must include a **public link to the conversation(s)** used. AUC's academic integrity rules apply: submitted work must reflect your own understanding.

Tasks:

Part 1: Model Training (15%)

Train one interpretable model:

- Decision Tree (max depth ≤ 5 recommended)
- Logistic Regression (optionally with L1 regularization)

Answer:

- What model did you choose?
- What is its performance? (could be same metrics as Module 1)

Hint: You could reuse your Module 1 pipeline. Only change the model if needed.

Part 2: Why is this model interpretable? (5%)

Explain why your model is interpretable by design. Relate your explanation to how a human can understand individual predictions.

Hint: Keep it conceptual (no code required here)

Part 3: What features matter the most? (15%)

Identify the most important features driving predictions.

Possible approaches:

- Decision Tree → feature importance (model.feature_importances_)
- Logistic Regression → coefficients (model.coef_)

Answer:

- Top 5–10 most important features
- Why do they make sense (or not)?

Part 4: How do features affect predictions? (15%)

Analyze direction and behavior of key features.

Answer:

- Does each feature increase or decrease prediction?
- Describe whether the effect is increasing, decreasing, or varies across values (based on splits or coefficients)

Hints:

- *Logistic Regression → sign of coefficient (+ / −)*
- *Decision Tree → observe splits (thresholds)*

Part 5: Local Explanations (2 individuals) (15%)

Pick 2 different individuals from your dataset and explain predictions.

Answer for each individual:

- Prediction outcome
- Key features influencing the decision
- Why the model made this decision

Hints:

- *Decision Tree → trace the path (conditions)*
- *Logistic Regression → compute feature contributions:*
- *contribution ≈ feature × coefficient*
- *Example: "Rejected because income low + debt high"*

Part 6: Detect Suspicious Patterns (20%)

Identify any questionable, unexpected, or unfair patterns.

Answer:

- Does anything look wrong or counterintuitive?
- Are any features behaving strangely?
- Could there be bias?

Hints:

- *Features strongly tied to protected attributes*
- *Unexpected relationships (e.g., higher income → worse outcome)*
- *Compare with fairness findings from Module 1*
- *Example: "Model relies heavily on race → potential bias"*

Part 7: Design Recommendation (15%)

Evaluate your system as if deploying it.

Answer:

- Would you use this model in a real system? Why or why not?
- What are the risks?
- Would you change anything?